

Power BI Interview Questions and Answers

1. What is Power BI?

Interviewer: What is Power BI?

Candidate: Power BI is a Business Intelligence and Data Visualization tool developed by Microsoft. It helps organizations connect to different data sources, clean and transform data, create interactive dashboards and reports, and share insights with stakeholders. In my projects, I have used Power BI to analyze business data and present it through interactive visualizations, making it easier for decision-makers to understand trends and performance.

2. What are the main components of Power BI?

Interviewer: Can you explain the main components of Power BI?

Candidate: Yes. The main components are:

- Power BI Desktop
- Power BI Service
- Power BI Mobile
- Power BI Gateway
- Power BI Report Server

Power BI Desktop is used to build reports, Power BI Service is used to publish and share reports online, Power BI Mobile allows users to view reports on mobile devices, Gateway refreshes on-premises data, and Report Server hosts reports within an organization's infrastructure.

3. What is Power BI Desktop?

Interviewer: What is Power BI Desktop?

Candidate: Power BI Desktop is a free Windows application where we connect data, clean it using Power Query, create relationships, write DAX

calculations, and build reports and dashboards. After developing reports, we publish them to the Power BI Service.

4. What is Power BI Service?

Interviewer: What is Power BI Service?

Candidate: Power BI Service is the cloud platform where reports and dashboards are published, shared, and refreshed automatically. It allows collaboration among team members and provides features like scheduled refresh, workspace management, Row-Level Security, and dashboards.

5. What is Power Query?

Interviewer: What is Power Query?

Candidate: Power Query is the ETL tool inside Power BI. ETL stands for Extract, Transform, and Load. It is used to clean and prepare data before loading it into the data model. Common operations include removing duplicates, changing data types, splitting columns, merging tables, and handling missing values.

6. What is DAX?

Interviewer: What is DAX?

Candidate: DAX stands for Data Analysis Expressions. It is the formula language used in Power BI to create calculated columns, measures, and calculated tables. DAX helps perform calculations like totals, averages, percentages, rankings, and time intelligence.

7. What is the difference between Power Query and DAX?

Interviewer: Can you differentiate between Power Query and DAX?

Candidate: Yes.

- Power Query is used before loading data to clean and transform it.
- DAX is used after loading data to perform calculations and create business logic.

In simple terms, Power Query prepares the data, while DAX analyzes the data.

8. What is Data Modeling?

Interviewer: What is Data Modeling?

Candidate: Data Modeling is the process of creating relationships between different tables. A good data model improves report performance and makes DAX calculations easier. Usually, we create relationships between fact tables and dimension tables.

9. What are Fact and Dimension Tables?

Interviewer: Explain Fact and Dimension Tables.

Candidate: A Fact Table contains numerical business data such as sales, revenue, or profit.

A Dimension Table contains descriptive information like customer details, product names, or dates.

For example, the Sales table is a Fact Table, while Customer, Product, and Date tables are Dimension Tables.

10. What is Star Schema?

Interviewer: What is Star Schema?

Candidate: Star Schema is a data modeling technique where one central Fact Table is connected to multiple Dimension Tables. It improves query performance and simplifies report development. It is considered the best practice in Power BI.

11. What are Relationships in Power BI?

Interviewer: What are Relationships?

Candidate: Relationships connect tables using common columns. They allow Power BI to combine data from multiple tables correctly. Without relationships, reports may display incorrect results.

12. What are the types of Relationships?

Interviewer: What types of relationships are available?

Candidate: There are four types:

- One-to-One
- One-to-Many
- Many-to-One
- Many-to-Many

Among these, One-to-Many is the most commonly used relationship in Power BI projects.

13. What is Cardinality?

Interviewer: What is Cardinality?

Candidate: Cardinality defines how records in one table relate to records in another table. It helps Power BI understand whether one record matches one, many, or multiple records in another table.

14. What is Cross Filter Direction?

Interviewer: What is Cross Filter Direction?

Candidate: Cross Filter Direction controls how filters move between related tables. It can be Single Direction or Both Direction. Single Direction is

generally recommended because it improves performance and reduces ambiguity.

15. What is a Measure?

Interviewer: What is a Measure?

Candidate: A Measure is a dynamic calculation created using DAX. It is calculated at runtime based on the filters applied in the report.

For example:

- Total Sales
- Average Sales
- Profit Margin

Measures are commonly used in KPI cards and charts.

16. What is a Calculated Column?

Interviewer: What is a Calculated Column?

Candidate: A Calculated Column is created using DAX and is calculated for every row when the data is loaded or refreshed. Unlike measures, calculated columns consume storage because their values are stored in the data model.

17. Difference Between Measure and Calculated Column?

Interviewer: Differentiate between Measure and Calculated Column.

Candidate: A Measure is calculated dynamically based on filters and does not consume storage.

A Calculated Column is calculated once during data refresh and stores values in the model.

Measures are preferred whenever possible because they improve model performance.

18. What are Visualizations in Power BI?

Interviewer: What are Visualizations?

Candidate: Visualizations are graphical representations of data that help users understand business insights easily.

Examples include:

- Bar Chart
- Line Chart
- Pie Chart
- Table
- Matrix
- Card
- KPI
- Map
- Funnel Chart

Choosing the right visualization makes reports more meaningful.

19. What is a Dashboard?

Interviewer: What is a Dashboard?

Candidate: A Dashboard is a single-page view created in Power BI Service. It combines visuals from one or multiple reports, allowing business users to monitor important KPIs in one place.

20. What is the Difference Between Report and Dashboard?

Interviewer: Differentiate between Report and Dashboard.

Candidate: A Report can contain multiple pages with detailed analysis and is created in Power BI Desktop.

A Dashboard is a single-page summary created in Power BI Service using visuals pinned from reports.

21. What is a Slicer?

Interviewer: What is a Slicer?

Candidate: A Slicer is a visual filter that allows users to interactively filter report data.

For example, users can filter reports by Year, Product, Branch, or Region without changing the original dataset.

22. What is Drill Down?

Interviewer: What is Drill Down?

Candidate: Drill Down allows users to move from higher-level data to more detailed information.

For example:

Year → Quarter → Month → Day

This helps analyze data at different levels.

23. What is Drill Through?

Interviewer: What is Drill Through?

Candidate: Drill Through allows users to navigate from a summary page to a detailed report page related to a selected value.

For example, clicking on a specific product can open a page showing detailed sales information for that product.

24. What is Row-Level Security (RLS)?

Interviewer: What is Row-Level Security?

Candidate: Row-Level Security restricts data access for different users.

For example, if a company has Sales Managers in different regions, each manager can view only their own regional sales data.

This improves data security and confidentiality.

25. What is Power BI Gateway?

Interviewer: Why do we use Power BI Gateway?

Candidate: Power BI Gateway securely connects on-premises data sources with Power BI Service. It enables scheduled refresh without moving the database to the cloud.

26. What is Scheduled Refresh?

Interviewer: What is Scheduled Refresh?

Candidate: Scheduled Refresh automatically updates reports in Power BI Service at predefined intervals. This ensures users always see the latest available data without manual intervention.

27. What Data Sources Can Power BI Connect To?

Interviewer: Which data sources have you worked with?

Candidate: Power BI supports many data sources, including:

- Excel
- CSV Files
- SQL Server

- MySQL
- Oracle
- PostgreSQL
- SharePoint
- Azure SQL
- Web APIs
- Folder Data

In my projects, I have mainly worked with Excel, CSV files, and SQL databases.

28. What are KPIs?

Interviewer: What are KPIs?

Candidate: KPI stands for Key Performance Indicator. KPIs measure business performance against predefined goals.

Examples include:

- Total Sales
- Revenue
- Profit
- Customer Count
- Conversion Rate
- Growth Percentage

KPIs are usually displayed using Card or KPI visuals.

29. What are the Steps to Create a Power BI Report?

Interviewer: Explain your Power BI development process.

Candidate: My usual workflow is:

1. Connect to the data source.
2. Clean and transform data using Power Query.
3. Load data into Power BI.
4. Create relationships.
5. Write DAX measures.
6. Build interactive visualizations.
7. Apply formatting and filters.
8. Test the report.
9. Publish it to Power BI Service.
10. Configure scheduled refresh and sharing.

This workflow helps build accurate and user-friendly reports.

30. Why Should We Hire You as a Power BI Developer?

Interviewer: Why should we hire you for a Power BI role?

Candidate: As a fresher, I have built multiple end-to-end Power BI projects where I connected data from different sources, cleaned it using Power Query, created relationships, wrote DAX measures, and developed interactive dashboards. I understand the complete Power BI workflow, including publishing reports to Power BI Service. I am a quick learner, enjoy solving business problems with data, and I am confident that I can contribute effectively while continuously improving my technical skills.